Assignments for

Analog & Digital Electronics (BE03009011)

B.E. Semester 3 (Electrical)

Institute logo

Directorate of Technical Education, Gandhinagar, Gujarat

L. E. College of Engineering, Ahmedabad

Certificate

This is to certify that Mr./Ms. _____ _____ Enrollment No. _____ of B.E. Semester _____ Electrical Engineering of this Institute (GTU Code: 031) has satisfactorily completed the Assignment work for the subject **Analog and Digital Electronics (BE03009011)** for the academic year 202 - 202 .

Place: _____

Date: _____

Name and Sign of Faculty member

Head of the Department

Preface

Main motto of any laboratory / practical / assignment / field work is for enhancing required skills as well as creating ability amongst students to solve real time problem by developing relevant competencies in psycho-motor domain. By keeping in view, GTU has designed competency focused outcome-based curriculum for engineering degree programs where sufficient weightage is given to practical work. It shows importance of enhancement of skills amongst the students. It is must for effective implementation of competency focused outcome-based curriculum that every practical / assignment is keenly designed to serve as a tool to develop and enhance relevant competency required by the various industry among every student.

By solving these assignments students can try to develop a better understanding of the subject. This in turn enhances pre-determined outcomes amongst students.

Analog and Digital Electronics is the fundamental course which deals with both analog and digital electronic signals and their conversion from one to another. Analog electronics is the branch of electronics that deals with electronic systems and devices with continuous time signals. The term ‘analog or analogue’ describes a proportional relationship between a signal and an electrical quantity (voltage or current) that represents the signal. In analog electronics, two types of components are used to design the systems which are: the active elements such as diode, transistors, etc. and the passive elements such as resistors, capacitors, inductors, etc. Digital electronics systems are usually made from a combination of logic gates, often packaged in an integrated circuit (IC). The digital electronics uses binary logic function to perform operations. Digital electronics uses active components only. One of the most common applications of digital electronics is in computers.

The utmost care has been taken while preparing these assignments however always there is chances of improvement. Therefore, we welcome constructive suggestions for improvement and removal of errors if any.

Practical – Course Outcome matrix

	Course Outcomes (COs): 1. Comprehension of various types of analog and digital components – OPAMP, Combinational and sequential circuits, ADC and DAC, involved in the analog and digital systems. 2. Describe the functioning and selection of OP-AMP as per application. 3. Design and testing of OP-AMP based circuits. 4. Design and implement Combinational and Sequential logic circuits. 5. Describe the process of Analog to Digital conversion and Digital to Analog conversion.					
Sr. No.	**Objective(s) of Experiment**	**CO 1**	**CO 2**	**CO 3**	**CO 4**	**CO 5**
1.	Assignment No. 1	√	√			
2.	Assignment No. 2	√		√		
3.	Assignment No. 3	√			√	
4.	Assignment No. 4	√				√

Industry Relevant Skills

The following industry relevant competency is expected to be developed in the student by undertaking the practical work of this laboratory.

1. An aptitude for math
2. Critical thinking skills
3. Knowledge of ICs

Note:

The theory, questions and figures used in this manual are used from the prescribed texts and references mentioned in the GTU syllabus:

1. Ramakant A Gayakwad, Op- Amps and Linear Integrated Circuits, Prentice Hall of India
2. A. Kumar, "Fundamentals of Digital Circuits", Prentice Hall India, 2016.
3. Website : nptel.ac.in
4. Google sites

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(Progressive Assessment Sheet)

Sr. No.	Assignment	Page No.	Date of performance	Date of submission	Assessment Marks	Sign. of Teacher with date	Remarks
Total							

Date:

Assignment No: 1

Relevant CO:

[CO-1] Comprehension of various types of analog and digital components – OPAMP, Combinational and sequential circuits, ADC and DAC, involved in the analog and digital systems

[CO - 2] Describe the functioning and selection of OP-AMP as per application.

Objectives: To understand how to

a. read a typical opamp data sheet,
b. properly interpret the data sheet
c. understand the characteristics of the opamp
d. Select the proper opamp for a desired application.

1. Explain the schematic block diagram of the basic op amp with suitable diagram.

2. Enlist and explain six characteristic of the ideal op amp.

3. Determine the output voltage Vo of the op amp for non-inverting input voltage Vi of 200 μV and inverting input voltage of 100 μV. The amplifier has a differential gain A_d of 1000 and CMRR: (1) 100, (2) 10^4

4. An op amp inverting amplifier has an input resistor of 10 KΩ and a feedback resistor of 50 KΩ. If the input voltage is 0.5 V, find the output voltage and input current.

5. If V_1 and V_2 are two voltages (with respect to ground), how would you construct an op amp circuit to get the voltage $Vo = 2V_1 – V_2$

Suggested References:

1. Ramakant A Gayakwad, Op-Amps and Linear Integrated Circuits, Prentice Hall of India
2. Website : nptel.ac.in

Rubric wise marks obtained:

01-Scope for improvement, 02-Moderate, 03-Good, 04-Very good, 05-Excellent					
Rubrics	**Level of Understanding**	**Analyzing skills**	**Writing skills**	**Submission in time**	**Total**

Date:

Assignment No: 2

Relevant CO:

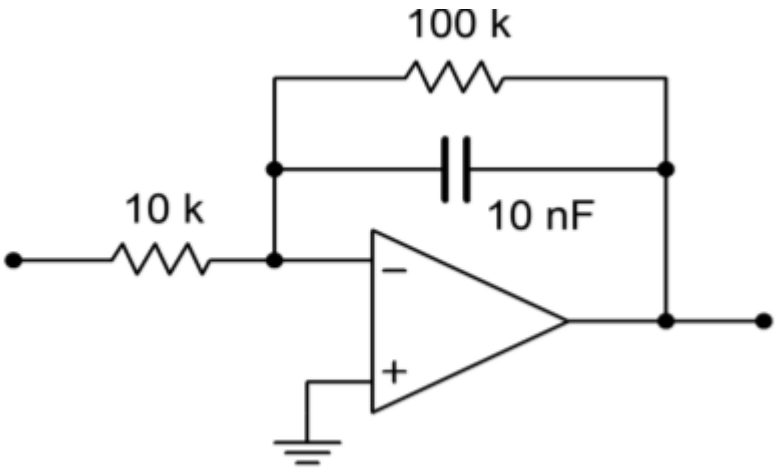
[CO-1] Comprehension of various types of analog and digital components – OPAMP, Combinational and sequential circuits, ADC and DAC, involved in the analog and digital systems

[CO - 3] Design and testing of OP-AMP based circuits.

Objectives: To understand how to

a. calculate parameters of the opamp
b. Design a desired application.

1. Determine the Equation for Vout, and the lower frequency limit of integration for the circuit:



2. Design an op-amp integrator circuit using a resistor R = 10 kΩ and a capacitor C = 1 μF. Apply a constant input voltage of Vin = 5V. Calculate the output voltage Vout after 1ms of operation

3. Design an op-amp differentiator circuit using R = 10kΩ and C=1μF. Apply a triangular wave input signal with a slope of 1000 V/s. Determine the output voltage Vout of the differentiator during the linear ramp portion of the triangle wave.

4. Explain Schmitt Trigger circuit using opamp with necessary waveforms.

5. Explain Precision rectifier and Hysteresis Comparator with necessary diagrams.

Suggested References:

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Rubrics	**Level of Understanding**	**Analyzing skills**	**Writing skills**	**Submission in time**	**Total**

Date:

Assignment No: 3

Relevant CO:

[CO-1] Comprehension of various types of analog and digital components – OPAMP, Combinational and sequential circuits, ADC and DAC, involved in the analog and digital systems

[CO - 4] Design and implement Combinational and Sequential logic circuits.

Objectives: To understand how to

a. properly interpret the variables
b. understand the combinational and sequential logic circuits.

1. Given the Boolean function: $f(A, B, C) = \Sigma m(1, 3, 5, 6)$

   Convert this function into Product of Maxterms form.

2. Given the Boolean function: $f(A, B, C) = \Pi M(1, 4, 6)$

   Convert this function into Sum of Minterms form.

3. Explain Master Slave J-K Flip Flop with necessary diagrams and truth tables.

4. Explain Universal shift register with necessary diagrams.

5. Reduce using K map and find out the minimal expression in SOP as well as POS logic:

   a) $f = \Sigma\, m(1, 2, 3, 9, 12, 13, 14)$

   b) $f = \Pi\, M(4, 5, 6, 7, 12, 13, 14, 15)$

   c) $f = \Sigma\, m(1,5,6,10,13,14) + d(2,4)$

   d) $f = \Pi\, M(6,7,8,9) \bullet d(10,11,12,13,14,15)$

Suggested References:

1. R. P. Jain, "Modern Digital Electronics", McGraw Hill Education, 2009.
2. M. M. Mano, "Digital logic and Computer design", Pearson Education India, 2016.
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Rubrics	**Level of Understanding**	**Analyzing skills**	**Writing skills**	**Submission in time**	**Total**

Date:

Assignment No: 4

Relevant CO:

[CO-1] Comprehension of various types of analog and digital components – OPAMP, Combinational and sequential circuits, ADC and DAC, involved in the analog and digital systems

[CO - 5] Describe the process of Analog to Digital conversion and Digital to Analog Conversion.

Objectives: To understand how to

a. Properly understand parameters of A/D and D/A converter
b. Design the A/D and D/A converter systems.

1. Design and analyze a Digital-to-Analog Converter (DAC) system.

Given: A 4-bit DAC (i.e., digital input has 4 bits). Reference voltage range: 0 to 10 V Digital input: 1101. Determine: (1) Determine the resolution of the DAC in volts. (2) Find the analog output voltage for the given digital input. (3) Explain the working of a DAC and mention some practical applications.

2. Explain the working Design and analyze an A/D converter system:

Given data: An analog input voltage range of 0 to 5 V. A 3-bit ADC (i.e., digital output has 3 bits). Determine (1) The resolution of the ADC in volts. (2) For an analog input of 3.2 V, find the corresponding digital output. (3) Explain the working of the ADC and its practical significance.

3. Explain different section of a ramp type A/D converter circuit.

4. List the different section of a counter ramp A/D converter circuit.

Suggested References:

1. R. P. Jain, "Modern Digital Electronics", McGraw Hill Education, 2009.
2. M. M. Mano, "Digital logic and Computer design", Pearson Education India, 2016.
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Rubrics	**Level of Understanding**	**Analyzing skills**	**Writing skills**	**Submission in time**	**Total**

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